

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS: 15 Total Marks: 25 Annexure A

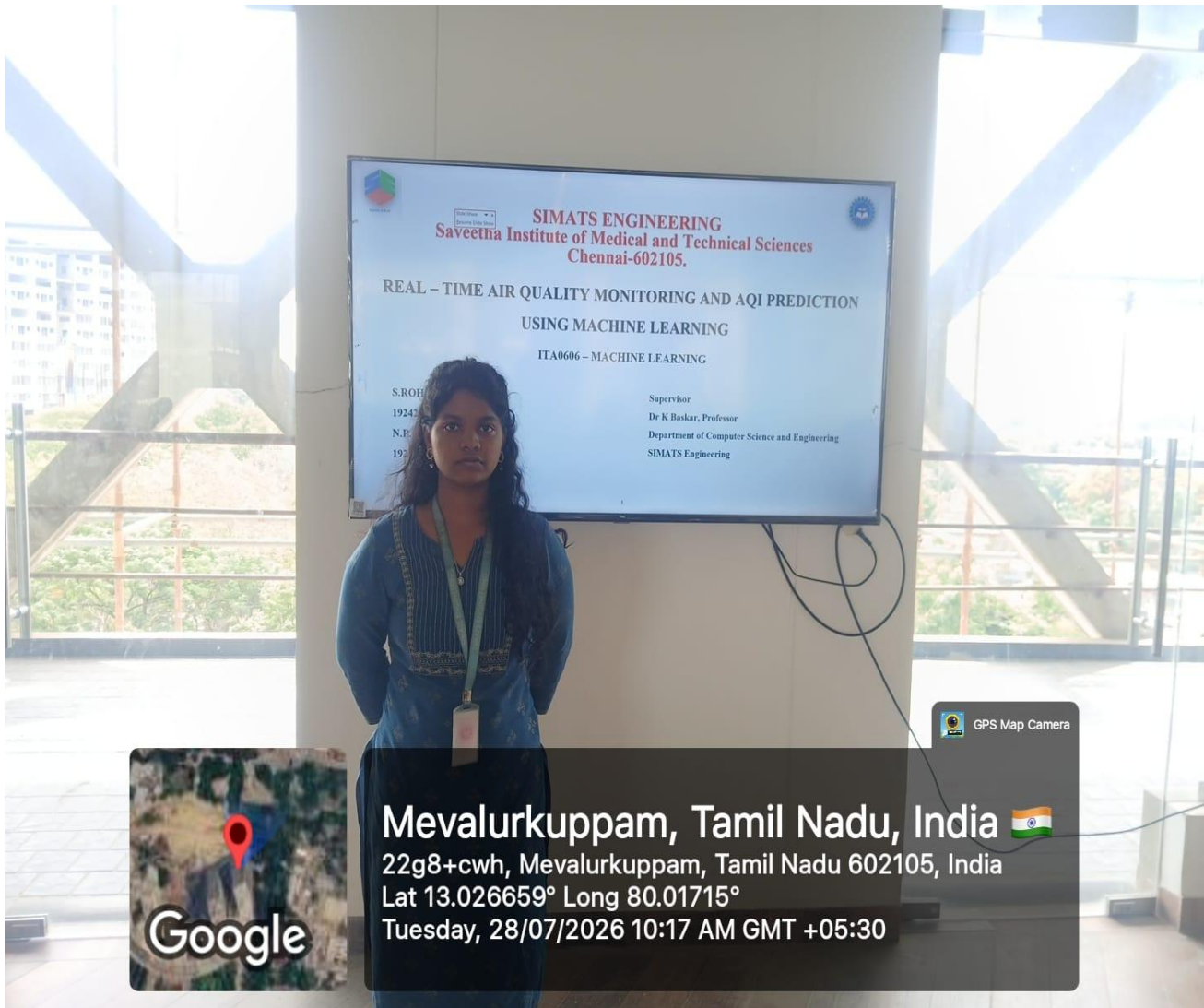
TECHNICAL PRESENTATION

Code of Conduct:

*I **Gayathri N P (192421362)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

GEOTAG PHOTO





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QUALITY MONITORING AND AQI PREDICTION REAL – TIME AIR USING MACHINE LEARNING ITA0606 – MACHINE LEARNING

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Problem Statement

- Current air quality monitoring systems primarily display real-time pollutant values without accurately predicting future Air Quality Index (AQI), limiting proactive environmental decision-making.
- Rapid changes in pollution levels due to traffic, industrial emissions, and weather conditions make it difficult for users to assess upcoming air quality and potential health risks.

This results in limited air quality awareness, delayed preventive actions, and increased health risks, especially for children, elderly people, and individuals with respiratory diseases.

Where Does the Problem Occur?

- Urban areas with high traffic congestion and industrial activities.
- Regions experiencing rapid changes in weather and pollution levels.
- Locations where continuous air quality monitoring and early AQI prediction are required.



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ABSTRACT

- The objective of this project is to develop a web-based system for Realtime air quality monitoring and Air Quality Index (AQI) prediction using machine learning. The system collects live environmental data, including PM2.5, PM10, CO, NO₂, SO₂, O₃, temperature, humidity, and wind speed, through an Air Quality API.
- The collected data is preprocessed and used as input to a Random Forest Regression model trained on historical air quality data to predict the AQI.
- The predicted AQI is classified into standard air quality categories, and appropriate health recommendations are provided to users.
- The expected result is an interactive Streamlit web application that accurately predicts AQI, displays real-time pollutant information, visualizes air quality trends, and helps users make informed decisions to reduce health risks.



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Objectives

To develop a web-based application that collects, monitors, and displays real-time air quality data, including PM2.5, PM10, CO, NO₂, SO₂, O₃, temperature, humidity, and wind speed, using an Air Quality API.

To predict the Air Quality Index (AQI) using the Random Forest Regression algorithm, classify air quality into standard AQI categories, and provide appropriate health recommendations through an interactive dashboard.



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Sl. No	Year of Publication	Authors	Title of the Publication	Findings	Limitations
1	2024	Mulomba Mukendi Christian, Hyebyong Choi	Air Quality Forecasting Using Machine Learning: A Global Perspective with Relevance to Low-Resource Settings	Random Forest provided reliable AQI prediction, especially for classification tasks in low-resource settings.	Performance depends on the availability and quality of environmental data.
2	2024	Kamaljeet Kaur Sidhu et al.	Predictive Modelling of Air Quality Index (AQI) Across Diverse Cities and States of India Using Machine Learning	Random Forest outperformed several ML models for AQI prediction using CPCB data.	Model focused on selected regions and required continuous data updates.
3	2025	Yildirim Öztürk, Feyyaz Alpaalaz, Emrah Aslan	Air Quality Forecasting Using Machine Learning: Comparative Analysis and Ensemble Strategies for Enhanced Prediction	Ensemble machine learning techniques improved AQI forecasting accuracy.	Increased computational complexity and training time.
4	2025	Emine Cengil	The Power of Machine Learning Methods and PSO in Air Quality Prediction	Random Forest and optimized ML methods achieved high prediction performance.	Optimization increased model complexity and computational cost.
5	2026	Xiaofeng Zhu et al.	Applicability Analysis of Tree-Based Ensemble Learning for Air Pollutant Prediction Models	Random Forest achieved excellent prediction accuracy for PM _{2.5} and PM ₁₀ and offered interpretable results.	Study was validated using data from specific cities and may require adaptation for other regions.

